Exercise sheet 1

Logic Programming (621.315), Winter semester 2026

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Quiz

1 *Each positive logic program has at most one model.*

Take $P = \{a :- b\}$. Both $\varnothing$ and $\{a, b\}$ satisfy the rule; they are distinct models. □

False

2 *Each positive logic program has at most one answer set.*

A positive program has a unique least model, obtained by iterating its consequence operator. Its answer set is this least model. □

True

3 *Each normal logic program has at least one model.*

Under the course definition, every rule has an atom in its head (no headless constraints). The interpretation $A(P)$ containing all atoms satisfies every rule, since every head is true. □

True

4 *Each normal logic program has at least one answer set.*

Take $P = \{a :- not\, a\}$. For $X = \varnothing$, the reduct contains the fact a and has least model $\{a\}$. For $X = \{a\}$, the reduct is empty and has least model $\varnothing$. Neither candidate equals its reduct least model. □

False

Exercise 1.1

Determine the answer set of the following positive logic programs.

(a)

Rule numbers refer to the following order (one rule ends at each period).

```
a.
b :- e.
b :- f.
c :- b, f.
c :- d, g.
d.
e :- b, d.
g :- a, b.
g :- a, d.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{d}\} \qquad \text{fact, rule 6}$$

$$M_3 = M_2 \cup \{\texttt{g}\} \qquad \text{rule 9}$$

$$M_4 = M_3 \cup \{\texttt{c}\} \qquad \text{rule 5}$$

No rule has a true body and a head outside M_4; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{g}\}$$

Neither b nor e has an initial derivation; f has no head rule. The rules linking b, e cannot start.

$$\boxed{X = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{g}\}}$$

(b)

Rule numbers refer to the following order (one rule ends at each period).

```
likes(kate,fish).
likes(kate,soup).
likes(mike,beef).
likes(mike,cake).
likes(paul,duck).
likes(paul,pork).
likes(tina,beef).
likes(tina,soup).
order(beef) :- likes(mike,beef), likes(tina,beef).
order(soup) :- likes(kate,soup), likes(tina,soup).
happy(kate) :- likes(kate,fish), order(fish).
happy(kate) :- likes(kate,soup), order(soup).
happy(mike) :- likes(mike,beef), order(beef).
happy(mike) :- likes(mike,cake), order(cake).
happy(paul) :- likes(paul,duck), order(duck).
happy(paul) :- likes(paul,pork), order(pork).
happy(tina) :- likes(tina,beef), order(beef).
happy(tina) :- likes(tina,soup), order(soup).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{likes(kate,fish)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{likes(kate,soup)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{likes(mike,beef)}\} \qquad \text{fact, rule 3}$$

$$M_4 = M_3 \cup \{\texttt{likes(mike,cake)}\} \qquad \text{fact, rule 4}$$

$$M_5 = M_4 \cup \{\texttt{likes(paul,duck)}\} \qquad \text{fact, rule 5}$$

$$M_6 = M_5 \cup \{\texttt{likes(paul,pork)}\} \qquad \text{fact, rule 6}$$

$$M_7 = M_6 \cup \{\texttt{likes(tina,beef)}\} \qquad \text{fact, rule 7}$$

$$M_8 = M_7 \cup \{\texttt{likes(tina,soup)}\} \qquad \text{fact, rule 8}$$

$$M_9 = M_8 \cup \{\texttt{order(beef)}\} \qquad \text{rule 9}$$

$$M_{10} = M_9 \cup \{\texttt{order(soup)}\} \qquad \text{rule 10}$$

$$M_{11} = M_{10} \cup \{\texttt{happy(kate)}\} \qquad \text{rule 12}$$

$$M_{12} = M_{11} \cup \{\texttt{happy(mike)}\} \qquad \text{rule 13}$$

$$M_{13} = M_{12} \cup \{\texttt{happy(tina)}\} \qquad \text{rule 17}$$

No rule has a true body and a head outside M_{13}; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{array}{c} \texttt{happy(kate)},\ \texttt{happy(mike)},\ \texttt{happy(tina)}, \\ \texttt{likes(kate,fish)},\ \texttt{likes(kate,soup)}, \\ \texttt{likes(mike,beef)},\ \texttt{likes(mike,cake)}, \\ \texttt{likes(paul,duck)},\ \texttt{likes(paul,pork)}, \\ \texttt{likes(tina,beef)},\ \texttt{likes(tina,soup)},\ \texttt{order(beef)}, \\ \texttt{order(soup)} \end{array} \right\}$$

$$X = \left\{ \begin{array}{c} \texttt{happy(kate)},\ \texttt{happy(mike)},\ \texttt{happy(tina)}, \\ \texttt{likes(kate,fish)},\ \texttt{likes(kate,soup)}, \\ \texttt{likes(mike,beef)},\ \texttt{likes(mike,cake)}, \\ \texttt{likes(paul,duck)},\ \texttt{likes(paul,pork)}, \\ \texttt{likes(tina,beef)},\ \texttt{likes(tina,soup)},\ \texttt{order(beef)}, \\ \texttt{order(soup)} \end{array} \right\}$$

Exercise 1.2

Determine the reducts for the given candidate sets and decide which candidates are answer sets.

(a)

Rule numbers refer to the following order (one rule ends at each period).

```
a :- b.
b :- not c.
c :- not b.
d :- a, e.
e.
```

Candidate X_1

$$X = \{\texttt{a},\ \texttt{b},\ \texttt{d},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 2 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
a :- b.
b.
d :- a, e.
e.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{fact, rule 4}$$

$$M_3 = M_2 \cup \{\texttt{a}\} \qquad \text{rule 1}$$

$$M_4 = M_3 \cup \{\texttt{d}\} \qquad \text{rule 3}$$

No rule has a true body and a head outside M_4; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{b},\ \texttt{d},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Candidate X_2

$$X = \{\texttt{b},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 2 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
a :- b.
b.
d :- a, e.
e.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{fact, rule 4}$$

$$M_3 = M_2 \cup \{\texttt{a}\} \qquad \text{rule 1}$$

$$M_4 = M_3 \cup \{\texttt{d}\} \qquad \text{rule 3}$$

No rule has a true body and a head outside M_4; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{b},\ \texttt{d},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) \neq X$$

$$\boxed{X \text{ is not an answer set.}}$$

Missing from the candidate: $\{\texttt{a},\ \texttt{d}\}$.

(b)

Rule numbers refer to the following order (one rule ends at each period).

```
a :- not b.
a :- d.
b :- not c.
c :- a, not b.
d :- a.
d :- not c, not e.
e :- b, not a.
e :- d.
```

Candidate X_1

$$X = \{\texttt{a},\ \texttt{b},\ \texttt{d},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 6 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule 7 is deleted because $\{\texttt{a}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
a :- d.
b.
d :- a.
e :- d.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

No rule has a true body and a head outside M_1; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b}\}$$

$$\mathrm{Cn}(P^X) \neq X$$

$$\boxed{X \text{ is not an answer set.}}$$

Unsupported candidate atoms: $\{\texttt{a},\ \texttt{d},\ \texttt{e}\}$.

Candidate X_2

$$X = \{\texttt{b},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 6 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule 7 is retained because $\{\texttt{a}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
a :- d.
b.
d :- a.
e :- b.
e :- d.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{rule 4}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Exercise 1.3

Find all answer sets of both programs from Exercise 1.2 without testing all interpretations.

(a)

The fact forces e. If $b \in X$, the rule for c is deleted, so c has no supporting rule. Thus $c \notin X$. The reduct contains the fact b; a follows from b, and d follows from a, e. This gives $X = \{a, b, d, e\}$. If $b \notin X$, the reduct contains the fact c. The rule for b is then deleted. Neither a nor d can be derived, giving $X = \{c, e\}$. These two cases exhaust membership of b. □

$$\boxed{\#\,\mathrm{AS}(P) = 2}$$

Answer set 1 and reduct check

$$X = \{\texttt{a},\ \texttt{b},\ \texttt{d},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 2 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
a :- b.
b.
d :- a, e.
e.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{fact, rule 4}$$

$$M_3 = M_2 \cup \{\texttt{a}\} \qquad \text{rule 1}$$

$$M_4 = M_3 \cup \{\texttt{d}\} \qquad \text{rule 3}$$

No rule has a true body and a head outside M_4; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{b},\ \texttt{d},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 2 and reduct check

$$X = \{\texttt{c},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 2 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{b}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
a :- b.
c.
d :- a, e.
e.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{c}\} \qquad \text{fact, rule 2}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{fact, rule 4}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{c},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

The complete list of answer sets is:

$$\boxed{X_1 = \{\texttt{a},\ \texttt{b},\ \texttt{d},\ \texttt{e}\}}$$

$$\boxed{X_2 = \{\texttt{c},\ \texttt{e}\}}$$

(b)

If $b \notin X$, the reduct contains the fact a. The rules for d, e, and c then derive these three atoms. Since $c \in X$, the rule for b is deleted. The only candidate in this case is $\{a, c, d, e\}$.

If $b \in X$, support for b requires $c \notin X$. Also $e \in X$: if $a \in X$, the rules d from a and e from d force e; if $a \notin X$, the rule e from $b, \mathit{not}\, a$ forces e. Therefore the rule with body $\mathit{not}\, c, \mathit{not}\, e$ is deleted. The remaining rules for a, d form only the positive cycle a from d, d from a, which has no initial fact. Neither atom belongs to the least model. We obtain $\{b, e\}$. These two cases exhaust membership of b. □

$$\boxed{\#\,\mathrm{AS}(P) = 2}$$

Answer set 1 and reduct check

$$X = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{b}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule 4 is retained because $\{\texttt{b}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is deleted because $\{\texttt{c},\ \texttt{e}\} \subseteq X$.

Rule 7 is deleted because $\{\texttt{a}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
a.
a :- d.
c :- a.
d :- a.
e :- d.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{c}\} \qquad \text{rule 3}$$

$$M_3 = M_2 \cup \{\texttt{d}\} \qquad \text{rule 4}$$

$$M_4 = M_3 \cup \{\texttt{e}\} \qquad \text{rule 5}$$

No rule has a true body and a head outside M_4; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 2 and reduct check

$$X = \{\texttt{b},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 6 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule 7 is retained because $\{\texttt{a}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
a :- d.
b.
d :- a.
e :- b.
e :- d.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{rule 4}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

The complete list of answer sets is:

$$\boxed{X_1 = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{e}\}}$$

$$\boxed{X_2 = \{\texttt{b},\ \texttt{e}\}}$$

Exercise 1.4

Determine all answer sets of the following normal logic programs.

(a)

Rule numbers refer to the following order (one rule ends at each period).

```
canary(tweety).
penguin(tux).
bird(tweety) :- canary(tweety).
bird(tux) :- penguin(tux).
injured(tweety) :- bird(tweety), not ok(tweety).
ok(tweety) :- bird(tweety), not injured(tweety).
injured(tux) :- bird(tux), not ok(tux).
ok(tux) :- bird(tux), not injured(tux).
flies(tweety) :- ok(tweety), not penguin(tweety).
flies(tux) :- ok(tux), not penguin(tux).
```

Both bird atoms follow from the facts. For each bird, at least one of `injured` and `ok` must hold: if neither held, the reduct would derive both from the bird atom. They cannot both hold, because then both supporting rules would be deleted. Thus each bird has exactly one of these two properties. The choices are independent. Tweety flies precisely when `ok(tweety)` is chosen, since `penguin(tweety)` has no supporting rule. The rule for Tux flying is always deleted by the penguin fact. The four combinations below exhaust the two independent binary choices. □

$$\boxed{\#\,\mathrm{AS}(P) = 4}$$

Answer set 1 and reduct check

$$X = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{flies(tweety)},\ \texttt{injured(tux)},\ \texttt{ok(tweety)}, \\ \texttt{penguin(tux)} \end{array} \right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 5 is deleted because $\{\texttt{ok(tweety)}\} \subseteq X$.

Rule 6 is retained because $\{\texttt{injured(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is retained because $\{\texttt{ok(tux)}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is deleted because $\{\texttt{injured(tux)}\} \subseteq X$.

Rule 9 is retained because $\{\texttt{penguin(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 10 is deleted because $\{\texttt{penguin(tux)}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
canary(tweety).
penguin(tux).
bird(tweety) :- canary(tweety).
bird(tux) :- penguin(tux).
ok(tweety) :- bird(tweety).
injured(tux) :- bird(tux).
flies(tweety) :- ok(tweety).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{canary(tweety)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{penguin(tux)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{bird(tux)}\} \qquad \text{rule 4}$$

$$M_4 = M_3 \cup \{\texttt{bird(tweety)}\} \qquad \text{rule 3}$$

$$M_5 = M_4 \cup \{\texttt{injured(tux)}\} \qquad \text{rule 6}$$

$$M_6 = M_5 \cup \{\texttt{ok(tweety)}\} \qquad \text{rule 5}$$

$$M_7 = M_6 \cup \{\texttt{flies(tweety)}\} \qquad \text{rule 7}$$

No rule has a true body and a head outside M_7; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{\begin{array}{c}\texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)},\\ \texttt{flies(tweety)},\ \texttt{injured(tux)},\ \texttt{ok(tweety)},\\ \texttt{penguin(tux)}\end{array}\right\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 2 and reduct check

$$X = \left\{\begin{array}{c}\texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)},\\ \texttt{flies(tweety)},\ \texttt{ok(tux)},\ \texttt{ok(tweety)},\ \texttt{penguin(tux)}\end{array}\right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 5 is deleted because $\{\texttt{ok(tweety)}\} \subseteq X$.

Rule 6 is retained because $\{\texttt{injured(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is deleted because $\{\texttt{ok(tux)}\} \subseteq X$.

Rule 8 is retained because $\{\texttt{injured(tux)}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is retained because $\{\texttt{penguin(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 10 is deleted because $\{\texttt{penguin(tux)}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
canary(tweety).
penguin(tux).
bird(tweety) :- canary(tweety).
bird(tux) :- penguin(tux).
ok(tweety) :- bird(tweety).
ok(tux) :- bird(tux).
flies(tweety) :- ok(tweety).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{canary(tweety)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{penguin(tux)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{bird(tux)}\} \qquad \text{rule 4}$$

$$M_4 = M_3 \cup \{\texttt{bird(tweety)}\} \qquad \text{rule 3}$$

$$M_5 = M_4 \cup \{\texttt{ok(tux)}\} \qquad \text{rule 6}$$

$$M_6 = M_5 \cup \{\texttt{ok(tweety)}\} \qquad \text{rule 5}$$

$$M_7 = M_6 \cup \{\texttt{flies(tweety)}\} \qquad \text{rule 7}$$

No rule has a true body and a head outside M_7; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{\begin{array}{c}\texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)},\\ \texttt{flies(tweety)},\ \texttt{ok(tux)},\ \texttt{ok(tweety)},\ \texttt{penguin(tux)}\end{array}\right\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 3 and reduct check

$$X = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{injured(tux)},\ \texttt{injured(tweety)},\ \texttt{penguin(tux)} \end{array} \right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 5 is retained because $\{\texttt{ok(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is deleted because $\{\texttt{injured(tweety)}\} \subseteq X$.

Rule 7 is retained because $\{\texttt{ok(tux)}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is deleted because $\{\texttt{injured(tux)}\} \subseteq X$.

Rule 9 is retained because $\{\texttt{penguin(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 10 is deleted because $\{\texttt{penguin(tux)}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
canary(tweety).
penguin(tux).
bird(tweety) :- canary(tweety).
bird(tux) :- penguin(tux).
injured(tweety) :- bird(tweety).
injured(tux) :- bird(tux).
flies(tweety) :- ok(tweety).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{canary(tweety)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{penguin(tux)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{bird(tux)}\} \qquad \text{rule 4}$$

$$M_4 = M_3 \cup \{\texttt{bird(tweety)}\} \qquad \text{rule 3}$$

$$M_5 = M_4 \cup \{\texttt{injured(tux)}\} \qquad \text{rule 6}$$

$$M_6 = M_5 \cup \{\texttt{injured(tweety)}\} \qquad \text{rule 5}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{injured(tux)},\ \texttt{injured(tweety)},\ \texttt{penguin(tux)} \end{array} \right\}$$

$$\mathrm{Cn}(P^X) = X$$

$\boxed{X \text{ is an answer set.}}$

Answer set 4 and reduct check

$$X = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{injured(tweety)},\ \texttt{ok(tux)},\ \texttt{penguin(tux)} \end{array} \right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 5 is retained because $\{\texttt{ok(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is deleted because $\{\texttt{injured(tweety)}\} \subseteq X$.

Rule 7 is deleted because $\{\texttt{ok(tux)}\} \subseteq X$.

Rule 8 is retained because $\{\texttt{injured(tux)}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is retained because $\{\texttt{penguin(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 10 is deleted because $\{\texttt{penguin(tux)}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
canary(tweety).
penguin(tux).
bird(tweety) :- canary(tweety).
bird(tux) :- penguin(tux).
injured(tweety) :- bird(tweety).
ok(tux) :- bird(tux).
flies(tweety) :- ok(tweety).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{canary(tweety)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{penguin(tux)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{bird(tux)}\} \qquad \text{rule 4}$$

$$M_4 = M_3 \cup \{\texttt{bird(tweety)}\} \qquad \text{rule 3}$$

$$M_5 = M_4 \cup \{\texttt{injured(tweety)}\} \qquad \text{rule 5}$$

$$M_6 = M_5 \cup \{\texttt{ok(tux)}\} \qquad \text{rule 6}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{injured(tweety)},\ \texttt{ok(tux)},\ \texttt{penguin(tux)} \end{array} \right\}$$

$$\mathrm{Cn}(P^X) = X$$

$\boxed{X \text{ is an answer set.}}$

The complete list of answer sets is:

$$\boxed{X_1 = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{flies(tweety)},\ \texttt{injured(tux)},\ \texttt{ok(tweety)}, \\ \texttt{penguin(tux)} \end{array} \right\}}$$

$$X_2 = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{flies(tweety)},\ \texttt{ok(tux)},\ \texttt{ok(tweety)},\ \texttt{penguin(tux)} \end{array} \right\}$$

$$X_3 = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{injured(tux)},\ \texttt{injured(tweety)},\ \texttt{penguin(tux)} \end{array} \right\}$$

$$X_4 = \left\{ \begin{array}{c} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{injured(tweety)},\ \texttt{ok(tux)},\ \texttt{penguin(tux)} \end{array} \right\}$$

(b)

Rule numbers refer to the following order (one rule ends at each period).

```
num(1).
num(2).
num(3).
lucky(1) :- num(1), not unlucky(1).
unlucky(1) :- num(1), not lucky(1).
lucky(2) :- num(2), not unlucky(2).
unlucky(2) :- num(2), not lucky(2).
lucky(3) :- num(3), not unlucky(3).
unlucky(3) :- num(3), not lucky(3).
p :- unlucky(1), not p.
p :- unlucky(2), not p.
p :- unlucky(3), not p.
```

For each number, exactly one of `lucky` and `unlucky` holds: neither would make both reduct rules fire, while both would delete both supporting rules. Suppose $p \in X$. Every rule with head p is deleted by p, so the least reduct model does not contain p, a contradiction. Hence $p \notin X$. Each remaining rule for p now forbids its unlucky atom: an unlucky atom would derive p in the reduct. All three numbers must therefore be lucky. There is only one candidate. □

$$\boxed{\#\,\mathrm{AS}(P) = 1}$$

Answer set 1 and reduct check

$$X = \left\{ \begin{array}{c} \texttt{lucky(1)},\ \texttt{lucky(2)},\ \texttt{lucky(3)},\ \texttt{num(1)},\ \texttt{num(2)}, \\ \texttt{num(3)} \end{array} \right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 4 is retained because $\{\texttt{unlucky(1)}\} \cap X = \varnothing$; its default negations are removed.

Rule 5 is deleted because $\{\texttt{lucky(1)}\} \subseteq X$.

Rule 6 is retained because $\{\texttt{unlucky(2)}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is deleted because $\{\texttt{lucky(2)}\} \subseteq X$.

Rule 8 is retained because $\{\texttt{unlucky(3)}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is deleted because $\{\texttt{lucky(3)}\} \subseteq X$.

Rule 10 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 11 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 12 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.
Rule numbers refer to the following order (one rule ends at each period).

```
num(1).
num(2).
num(3).
lucky(1) :- num(1).
lucky(2) :- num(2).
lucky(3) :- num(3).
p :- unlucky(1).
p :- unlucky(2).
p :- unlucky(3).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{num(1)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{num(2)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{num(3)}\} \qquad \text{fact, rule 3}$$

$$M_4 = M_3 \cup \{\texttt{lucky(1)}\} \qquad \text{rule 4}$$

$$M_5 = M_4 \cup \{\texttt{lucky(2)}\} \qquad \text{rule 5}$$

$$M_6 = M_5 \cup \{\texttt{lucky(3)}\} \qquad \text{rule 6}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{matrix} \texttt{lucky(1)},\ \texttt{lucky(2)},\ \texttt{lucky(3)},\ \texttt{num(1)},\ \texttt{num(2)}, \\ \texttt{num(3)} \end{matrix} \right\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

The complete list of answer sets is:

$$\boxed{X_1 = \left\{ \begin{matrix} \texttt{lucky(1)},\ \texttt{lucky(2)},\ \texttt{lucky(3)},\ \texttt{num(1)},\ \texttt{num(2)}, \\ \texttt{num(3)} \end{matrix} \right\}}$$